

Supplementary Information for

DATA: Diafiltration Apparatus for high-Throughput Analysis

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S1. Model Derivation

Note that Fig. 1, Eq. (1), etc. refer to elements of the main text. Fig. S1, Eq. (S1), etc. are unique to the supplementary information.

In order to infer the three governing parameters in Eqs. (1) and (2), we conduct mass and solute balances around three control volumes within Fig. 1, the diafiltrate reservoir, the stirred cell, and the permeate. By assuming Fig. 1 represents a closed system, where the solution density is invariant to the salt concentration, the mass exiting the diafiltrate reservoir must be equal to the mass of water permeating through the membrane, Eq. (S1).

$$\frac{dm_d}{dt} = -A_m \rho J_w \quad (\text{S1})$$

where m_d is the mass of the diafiltrate reservoir, A_m is the membrane area and ρ is the density of the solution. As the diafiltrate concentration is defined before the experiment begins and nothing enters the diafiltrate reservoir, the diafiltrate concentration remains constant during each experiment, Eq. (S2).

$$\frac{dc_d}{dt} = 0 \quad (\text{S2})$$

Within our second control volume, the stirred cell, we enforce a constant volume system, i.e., Eq. (S3), by experimentally ensuring the volumetric flow rate of diafiltrate into the stirred cell is equal to the flow rate of the permeate across the membrane.

$$\frac{dm_f}{dt} = 0 \quad (\text{S3})$$

The solute within the stirred cell is modeled by finding the difference between the solute entering (i.e., from the diafiltrate) and the solute transported across the membrane, Eq. (S4). The left hand side of Eq. (S4) can be simplified by applying the product rule and substituting in Eq. (S3). This provides Eq. (5) which describes the changing concentration within the stirred cell.

$$\frac{d(c_f m_f)}{dt} = A_m \rho (J_w c_d - J_s) \quad (\text{S4})$$

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$$\frac{d(c_f m_f)}{dt} = m_f \frac{dc_f}{dt} + c_f \frac{dm_f}{dt} \quad (\text{S5})$$

$$\frac{dc_f}{dt} = \frac{A_m \rho}{m_f} (J_w c_d - J_s) \quad (5)$$

The final control volume encompasses the hold up volume, the transfer tube, and the scintillation vial. Equation (S6) states the permeate mass in this control volume remains constant during each experiment.

$$\frac{dm_h}{dt} = 0 \quad (\text{S6})$$

The amount of solute in the hold-up volume is provided by the difference of the solute transported through the membrane and the solute collected within the scintillation vial, Eq. (S7). Substituting Eq. (7) into Eq. (S7) yields Eq. (6).

$$\frac{d(c_h m_h)}{dt} = A_m \rho J_s - \frac{dm_v}{dt} c_h \quad (\text{S7})$$

$$\frac{dm_v}{dt} = -\frac{dm_d}{dt} = A_m \rho J_w \quad (7)$$

$$\frac{dc_h}{dt} = \frac{A_m \rho}{m_h} (J_s - c_h J_w) \quad (6)$$

The time dependent concentration of the scintillation vial is expressed by Eq. (8). As described previously, the product rule, Eq. (S8) is used to derive Eq. (S9) which captures the changing vial concentration.

$$\frac{d(c_v m_v)}{dt} = \frac{dm_v}{dt} c_h = A_m \rho J_w c_h \quad (8)$$

$$\frac{d(c_v m_v)}{dt} = m_v \frac{dc_v}{dt} + c_v \frac{dm_v}{dt} \quad (\text{S8})$$

$$\frac{dc_v}{dt} = \frac{dm_d}{dt} \frac{(c_v - c_h)}{m_v} = -A_m \rho J_w \frac{(c_v - c_h)}{m_v} \quad (\text{S9})$$

S2. Supplementary Figures

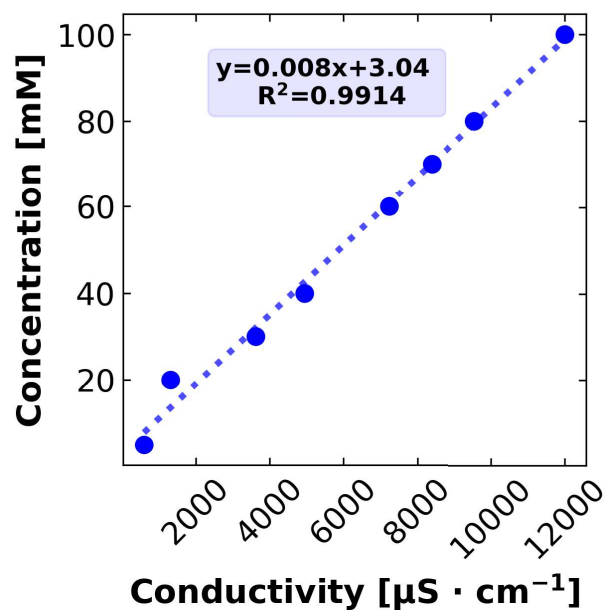


Fig. S1: The calibration curve used to convert conductivity measurements into the corresponding KCl concentration. The conductivity was calculated using, $C = GAV^{-1}$, the relationship between the conductivity probe cell constant ($G = 0.42 \text{ cm}^{-1}$), the peak to peak applied current, ($A = 1 \text{ mA}$), and the peak to peak differential voltage drop, V .

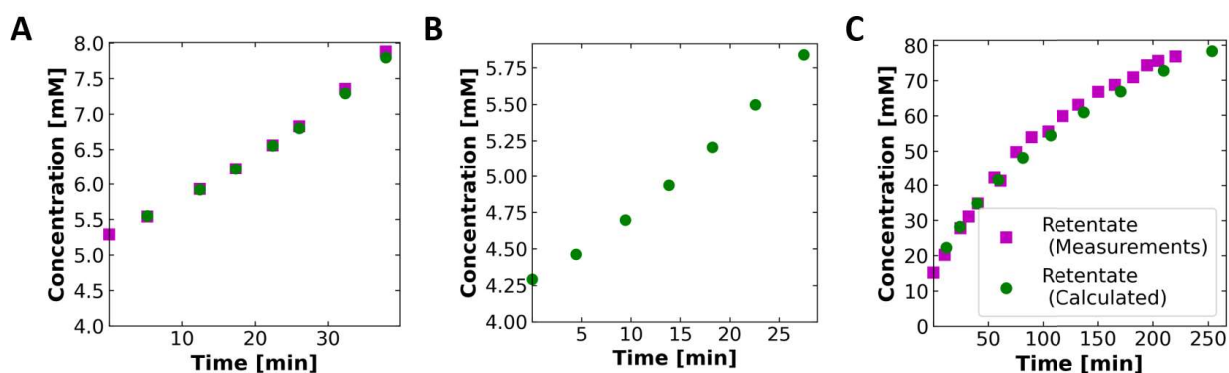


Fig. S2: Panel A presents a filtration experiment run at 40% recovery. The lower recovery allows the semi-continuous monitoring of the retentate with the inline conductivity probe (purple squares). Mass balance calculations (green circles) corroborate the conductivity measurements. Panel B presents the retentate concentration data, predicted by mass calculations (green circles), of a filtration experiment run at 60% recovery. The inline conductivity probe was not used as the high percent recovery decreased the volume of solution in the stirred cell below the sensor. Panel C presents information for a diafiltration experiment. The retentate concentration (purple squares) measured by the inline conductivity probe agrees with the the retentate concentration predicted from mass balance calculations (green circles). For filtration experiments, the calculated retentate concentration values were obtained by subtracting the total moles of solute in the permeate from the initial feed value and adjusting the stirred cell volume to account for the mass of solution in the permeate. Within diafiltration experiments, the stirred cell volume was kept constant. The change in solute within the stirred cell is calculated as the difference from the entering diafiltrate solute and exiting permeate solute. To calculate the new retentate concentration, the adjusted stirred cell solute amount is divided by the total volume.

Table S1: The time required to characterize a membrane over the phase space of interest (i.e., 5 mM - 80 mM) for filtration and diafiltration experiments are compared. Approximately 5 mL of DI water were run through the membranes before each experiment. Filtration experiments were run at 30% - 40% recovery with the inline conductivity probe; a total of 9 samples were collected from each experiment. Within the diafiltration experiments, an 80 mM diafiltrate concentration was used and a total of 13 samples were collected. The ICP-OES preparation and run time calculations were determined from the number of samples collected within the experiment. The analysis conducted consists of mass balances to ensure that all the solute entering the stirred cell was accounted for within the final retentate and permeate samples. While one diafiltration experiment covers the entire concentration range of interest, it would require approximately 10 filtration experiments to obtain an equal amount of information.

Filtration and Diafiltration Time Comparison			
		Filtration	Diafiltration
Avg. Samples Collected:	-	9	13
Avg. Permeated Mass:	gram	3.8	10.8
Experimental Set-up Time:	minute	20	20
Avg. Experiment Run Time:	minute	40	213
Membrane Rinse/Wash:	minute	60	60
Data Analysis (ICP) Prep. Time:	minute	36	52
ICP Run Time:	minute	67	79
Long Hand Analysis/Balances:	minute	60	60
Experimental Time:	minute	283	484
Phase Space Consideration:	experiment	10	1
Total Time:	minute	2830	484

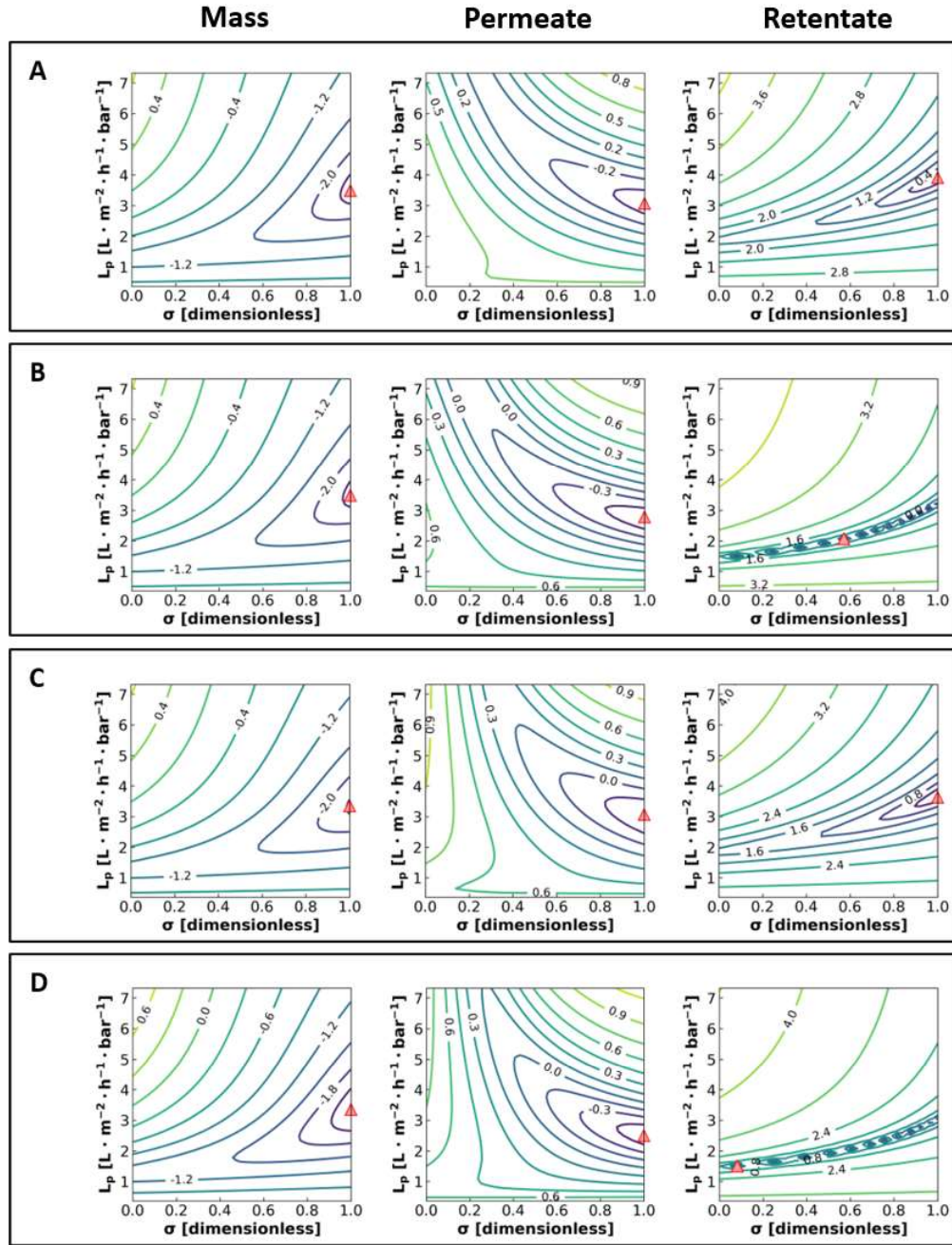


Fig. S3: Residual squared contours comparing the reflection coefficient and hydraulic permeability for **diafiltration** experiments. Two different models (with concentration polarization & without concentration polarization) and two data set variations (data with an inline conductivity sensor & data encompassing only the initial and final in-situ retentate measurements) are evaluated. The model used to generate the contours of panel A includes concentration polarization phenomena with the inline conductivity probe measurements. The residual squared contours of panel B uses a reduced data set which excludes the semi-continuous retentate data. Panel C contours evaluate a separate model in which the inline conductivity measurements are used yet concentration polarization is not accounted for. The contours generated in panel D use the reduced data set and neglect concentration polarization.

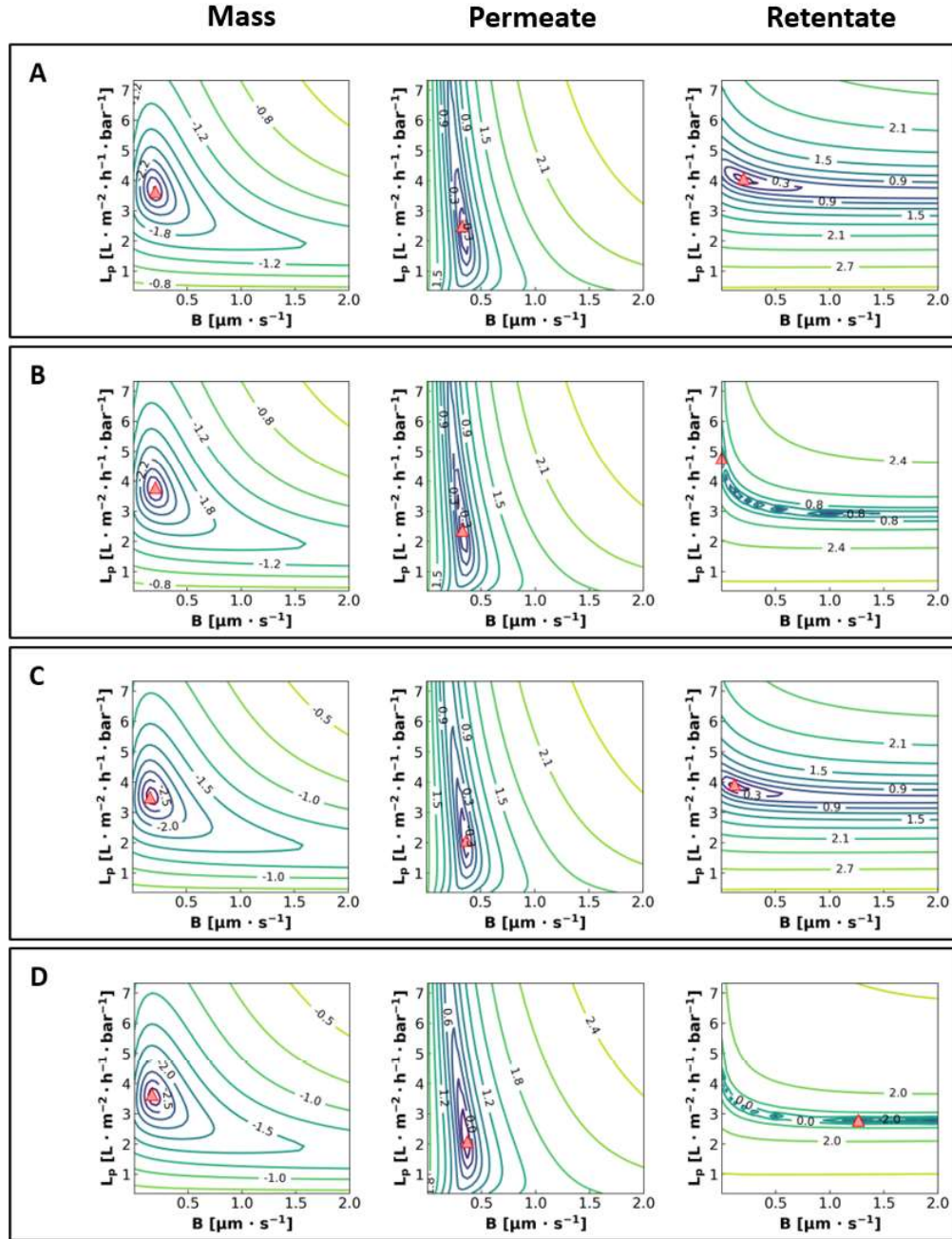


Fig. S4: Residual squared contours comparing the hydraulic permeability and solute permeability coefficient for **diafiltration** experiments. Two different models (with concentration polarization & without concentration polarization) and two data set variations (data with an inline conductivity sensor & data encompassing only the initial and final in-situ retentate measurements) are evaluated. The model used to generate the contours of panel A includes concentration polarization phenomena with the inline conductivity probe measurements. The residual squared contours of panel B uses a reduced data set which excludes the semi-continuous retentate data. Panel C contours evaluate a separate model in which the inline conductivity measurements are used yet concentration polarization is not accounted for. The contours generated in panel D use the reduced data set and do not incorporate concentration polarization.

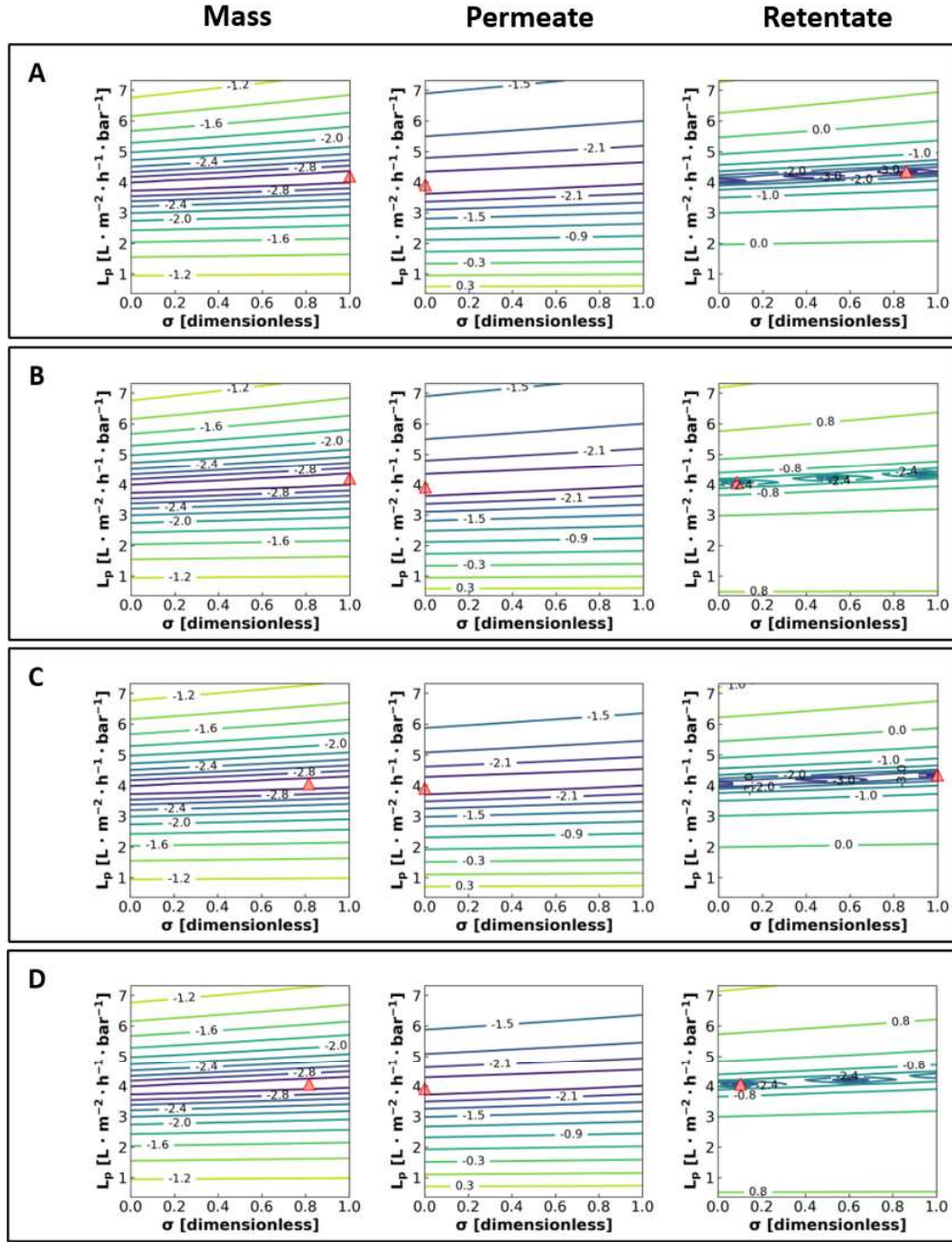


Fig. S5: Residual squared contours comparing the hydraulic permeability and reflection coefficient for of **filtration** experiments. Two different models (with concentration polarization & without concentration polarization) and two data set variations (data with an inline conductivity sensor & data encompassing only the initial and final in-situ retentate measurements) are evaluated. The model used to generate the contours of panel A includes concentration polarization phenomena with the inline conductivity probe measurements. The residual squared contours of panel B uses a reduced data set which excludes the semi-continuous retentate data. Panel C contours evaluate a separate model in which the inline conductivity measurements are used yet concentration polarization is not accounted for. The contours generated in panel D use the reduced data set and ignore concentration polarization.

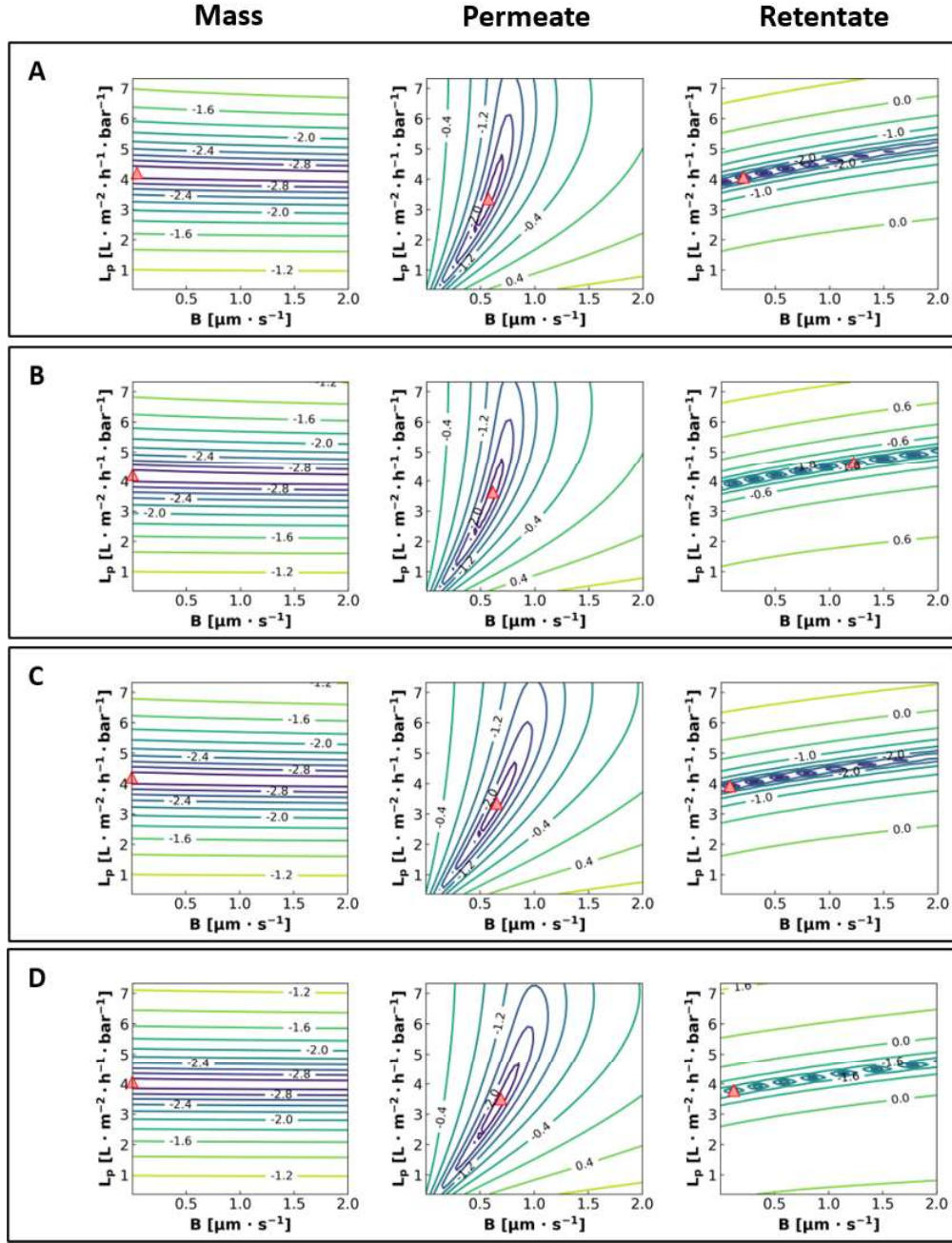


Fig. S6: Residual squared contours comparing the hydraulic permeability and solute permeability coefficient for of **filtration** experiments. Two different models (with concentration polarization & without concentration polarization) and two data set variations (data with an inline conductivity sensor & data encompassing only the initial and final in-situ retentate measurements) are evaluated. The model used to generate the contours of panel A includes concentration polarization phenomena with the inline conductivity probe measurements. The residual squared contours of panel B uses a reduced data set which excludes the semi-continuous retentate data. Panel C contours evaluate a separate model in which the inline conductivity measurements are used yet concentration polarization is not accounted for. The contours generated in panel D use the reduced data set and neglect concentration polarization.

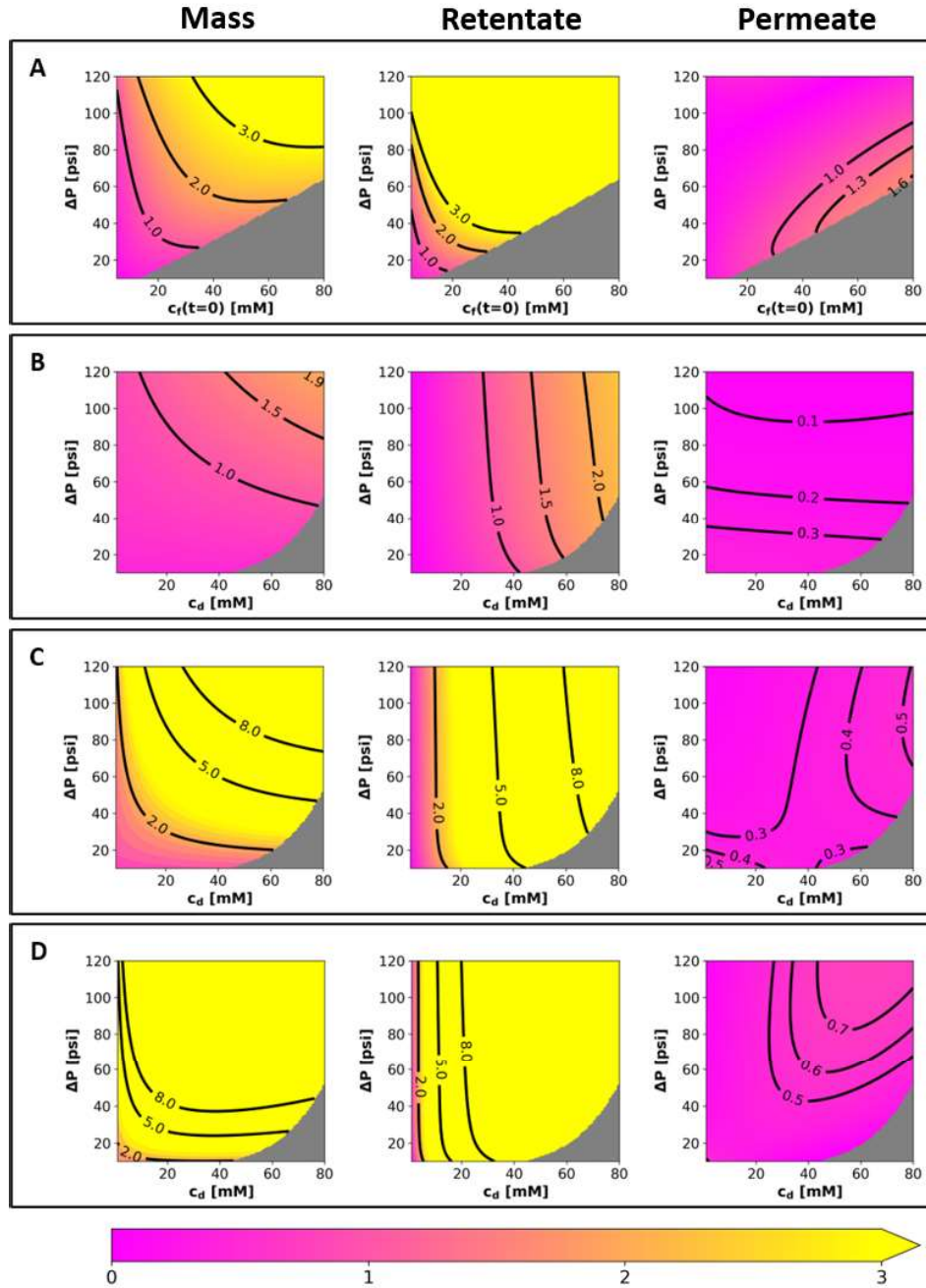


Fig. S7: Filtration and diafiltration experiments were modeled under varying initial feed or diafiltrate concentrations and applied pressures to determine the conditions necessary to differentiate reflection coefficients with a precision equal to 0.1. Panel A models a filtration experiment where the initial feed concentration and applied pressure are varied. Panels B, C, and D vary the applied pressure and diafiltrate concentration of a diafiltration experiment run with an initial feed concentration of 5mM. Panels B, C, and D correspond to the data gathered after the 1st, 5th, and 10th vials, respectively. The contours are generated using the L_p , B , σ values from Membrane A & M1 presented in Table 2. The contours represent the difference in model predictions via equations (7),(5), and (S9) from two different sigma values (i.e., $\sigma = 0.9$ and $\sigma = 1$) normalized by the precision of the measuring instruments. Thus the contour lines are dimensionless. The shaded area in the bottom right of the graphs correspond to non-physical systems in which the flux is equal to or less than zero (i.e., the osmotic pressure is equal to or greater than the applied pressure.)